Integrating “Power via Media Dependent Interface” and “Power over Data Line” support to the Linux kernel

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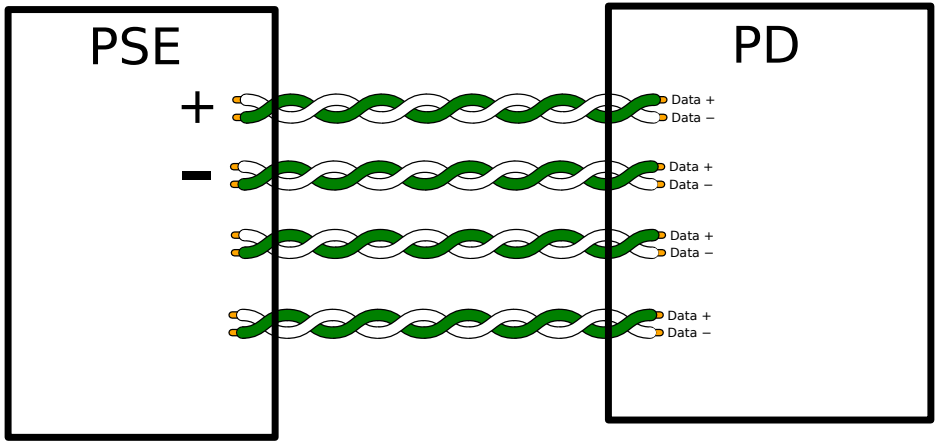
Patch v7: Add generic support for the PSE

- https://lore.kernel.org/all/20220926112500.990705-1-o.rempel@pengutronix.de/

What is Power via MDI?

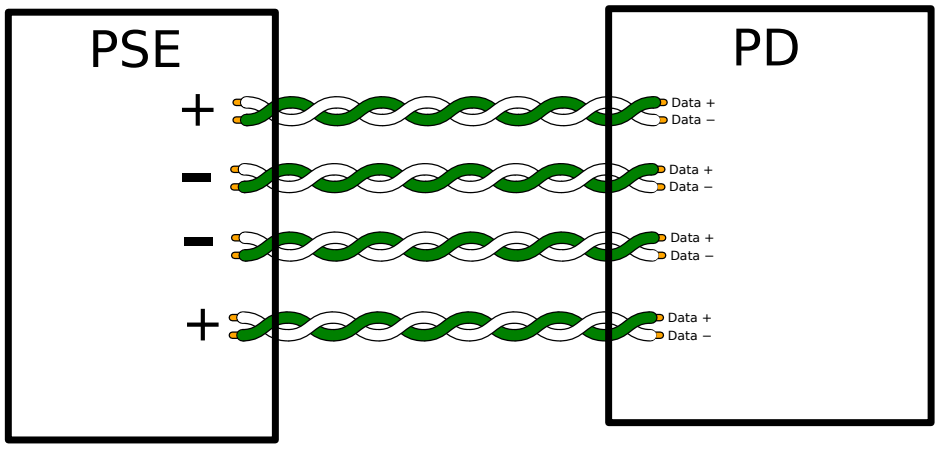
- Power via Media Dependent Interface (MDI) aka Power over Ethernet (PoE)
- Added with 802.3af-2003 and modified with 802.3at-2009
- PSE – power source equipment
- PD – powered device



- Power delivered over two pairs. One pair +, other pair -

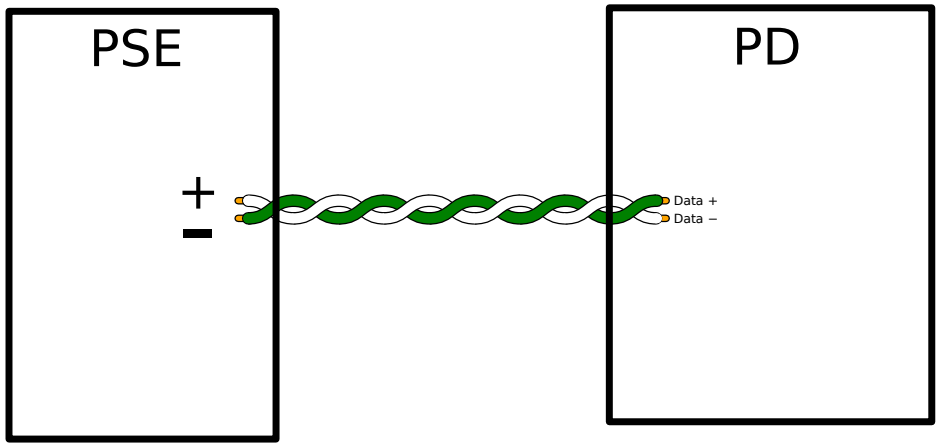
Extended to 4 pairs by 802.3bt

- Same as PoE but more power
- For power delivery are used all 4 pairs



What is PoDL?

- Power over Data Line
- For 10/100/1000Base-T1/L/S
- Same as PoE, except power is delivered over one twisted pair
- PoDL needs different filter implementation compared to PoE* variants



Detection

- Optional functionality
- PSE provides a constant current on the line and measures the voltage drop if PD is connected

Classification

- Optional functionality
- PoE and PoDL have different way for classification.
- PoDL is using SCCP – simple, digital, one-wire communication serial classification protocol.
- PoE uses a constant voltage from the PSE and a current sink on the PD to signal the power class

Challenges of kernel implementation

- How to name framework?
  - “Power over Ethernet” (PoE) is not used by the 802.3-2018 specification. Instead we have “Power via Media Dependent Interface” (PvMDI)
  - Power over Data Line (PoDL) is used by 802.3-2018 but not compatible with PvMDI
  - Amendment names (802.3af, 802.3at) do not scale and actual changes history is not preserved in finale spec.
  - PSE and PD are only common names which can be found in 802.3 specification for PvMDI and PoDL, and IC datasheets.

Challenges of kernel implementation

- How to name UAPI?
  - IEEE 802.3-2018 describes two types of PSE:
    - “PSE” for PvMDI/PoE*
    - “PoDL PSE” for PoDL
  - PSE and PoDL PSE have similar objects but different values and different standardization paths. Even if there are similarities now, they may become not compatible in the future.

IEEE 802.3-2018 to Kernel UAPI mapping

- aPoDLPSEAdminState

  → ETHTOOL_A_PODL_PSE_ADMIN_STATE

- aPoDLPSEPowerDetectionStatus

  → ETHTOOL_A_PODL_PSE_PW_D_STATUS

- acPoDLPSEAdminControl

  → ETHTOOL_A_PODL_PSE_ADMIN_CONTROL

Current kernel implementation

- Keep initial implementation minimal. Implement only what I really use and able to test.
- For now only PoDL PSE support
- PSE support on each network interface
- Can be detected from user space
- Provides ethtool interface
- Currently it can:
  - control admin state of PSE per port independent of link admin state
  - Read admin state and status of PSE

WIP: ethtool implementation

```
$ ip l
...
5: t1l1@eth0: <BROADCAST,MULTICAST> ..
...
$ ethtool --show-pse t1l1
PSE attributs for t1l1:
PoDL PSE Admin State: disabled
PoDL PSE Power Detection Status: disabled

$ ethtool --set-pse t1l1 podl-pse-admin-control enable
$ ethtool --show-pse t1l1
PSE attributs for t1l1:
PoDL PSE Admin State: enabled
PoDL PSE Power Detection Status: delivering power
```

What is next?

- Classification support.
  - Potential conflict with autoneg or manual master clock role (SCCP vs FLP)
- Components affecting maximal power delivery to the PD: Power supply, Board, Ethernet cable.
  - We need interfaces to set: max class, max load limit, load balancing, port prioritization
- Reuse existing frameworks where possibles: regulator, power delivery, etc.

What is next?

- PD Support
  - get classification result from PD controller
  - Report available power budget to related system components or to the user space
  - Possible use case: surveillance camera has enough power for CPU but no budget for the motor. Administrator will be able to get needed information for troubleshooting.

Thank you!

Questions?